

Project 20

Repeatable Closed-Path Motion on iiwa 7

Format: Group of two students; simulation-based project.

Prerequisites: Closed-path continuation, redundant inverse kinematics, null-space motion, and graph-based path planning.

Motivation

A redundant robot can return its tool to the starting pose while ending in a different posture. Repeating the same task can then accumulate joint drift or encounter new constraints. Repeatable motion requires closing the joint trajectory as well as the workspace loop.

Task

Overall objectives. Use classified iiwa IK feasibility charts to seek a redundancy schedule that closes in both full-pose workspace and joint space, and compare it with local continuation of the same closed task.

- **Prepare an explicitly closed pose loop.** Use the supplied rectangular path or a short derived loop and verify both translational and rotational closure. Add a correctly interpolated closing segment if needed; do not assume the first and last CSV rows are identical. Validate the model, tool transform, limits, and one admissible initial configuration.
- **Measure a local baseline.** Track the loop with the same pseudoinverse/null-space policy on every lap. Continue from the actual final configuration for up to three laps and record endpoint joint displacement, pose error, and feasibility. A closed tool pose alone does not establish joint-space repeatability.
- **Generate and classify the global chart.** Evaluate fixed- q_3 IK families over one full loop. Plot path sample versus redundancy angle, with branch labels and failure masks. Inspect the first and last columns as IK sets of the same pose, including their configuration correspondence.
- **Enforce joint-space closure.** Use a layered search with the initial configuration fixed and require the last configuration to match it within a declared tolerance and consistent permitted angle representatives. Validate the closing edge. Returning to the same redundancy angle or branch label alone is insufficient.
- **Compare repeated execution.** Refine the selected schedule and validate all transitions, including the seam between laps. Compare the local and closed schedules by endpoint joint error, travel, minimum limit margin, and full-Jacobian regularity. Show both trajectories on the feasibility chart and explain any accumulated posture drift.

Specifics and scope

Use one closed pose loop, one shared starting configuration, and at most three validation laps. Begin with about 150 pose samples and 120 redundancy angles. Search only within the selected sampled families; if no closing schedule is found, report the search limits rather than claiming global impossibility. Timing optimization is not required.

For every chart, keep pose validity, joint limits, full-Jacobian regularity, and reduced-chart regularity as separate diagnostics. The rectangular 6×7 Jacobian has no determinant. A feasible pixel is not a validated edge: re-solve intermediate prescribed poses and check joint continuity within limits. If collision checking is unavailable, state that feasibility is kinematic only; a clearance proxy is not a full collision test.

Expected outcome

Understand the distinction between tool-pose closure and repeatable robot motion. Deliver a classified redundancy chart with endpoint IK matches, a comparison of local and closure-constrained schedules, and lap-by-lap evidence of joint closure or drift.

Submit reproducible code, model parameters and test data, the required plots, a concise 5–6-page report, and a short simulation demonstration. Explain both successful cases and failures; conclusions must follow the numerical and geometric evidence rather than an expected outcome.

Resources

The KUKA LBR iiwa 7 robot description (URDF) and visualization meshes (STL) will be provided to students, together with a mathematical continuation model with unrestricted joints and an analytical fixed- q_3 IK model. Reuse the supplied IK rather than deriving a general 7R solver. Apply the physical joint limits from the robot description separately when using the unrestricted model. Check D–H parameters, joint axes, and the fixed tool transform against the URDF, and validate IK outputs by independent forward kinematics.

A sampled rectangular full-pose path and a reference fixed- q_3 IK table will also be provided. The pose data contain position coordinates and quaternion columns $\mathbf{qw}, \mathbf{qx}, \mathbf{qy}, \mathbf{qz}$; normalize the quaternions and make their signs consistent before interpolation. The reference IK table checks solutions at $q_3 = 30^\circ$ only; it does not replace evaluation at other angles. Use its configurations to check pose residuals and branch labels, applying physical joint limits separately.

Implement the local continuation baseline and a layered search through the redundant IK map using the supplied model and IK. Check workspace closure, match endpoint joint configurations, and validate the closing edge and inter-lap continuity before visualizing the trajectories with the supplied meshes.